

AI-POWERED REAL-TIME LOG MONITORING AND INCIDENT RESPONSE SYSTEM FOR WEB APPLICATIONS

ABSTRACT

Due to distributed systems, microservices, and continuous user traffic, modern web applications generate enormous amounts of log data, which leads to slow, imprecise, and ineffective log analysis techniques. Long downtime and low system reliability can result from late detection of critical events. The given paper suggests a real-time log monitoring and incident response system powered by AI implemented to monitor the web applications that will automate the process of log ingestion, analysis, and anomaly detection. The system applies anomaly detection methods that are anchored on machine learning and matches events among various application components. Through analyzing live log streams, the framework can identify possible failures in time, and alerts and remediation information can be taken through an interactive dashboard. The suggested solution lowers operational overhead, quickens incident reaction, and enhances general versatility of the applications with scalable and open-source innovations.

KEYWORDS : Log Monitoring, Anomaly Detection, Incident Response, Machine Learning, Real Time Analysis, Event Correlation

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